

Coding assignments

Coding assignments play a vital role in technical learning and evaluation. Beyond assessing coding ability, they foster critical thinking, problem-solving, and applied understanding. Here are six structured and compelling reasons why they matter:

01 Demonstrate real-world problem-solving

02 Reinforce core concepts via application

03 Handle raw, messy datasets

04 Showcase end-to-end project skills

05 Translate analysis into insights

06 Build confidence and industry readiness

WHAT DO THE ASSIGNMENTS ENTAIL?

The assignments are practical coding exercises derived from the materials that have been shared. They are intended to help candidates understand and develop practical coding skills. While a candidate's submitted work will be evaluated for its correctness and the quality of the code, the exercises are designed as opportunities for skill development, not as graded tests. Consequently, candidates will not receive a formal grade or marks. Instead, the focus is on providing helpful feedback to support their learning and improvement.

HOW CANDIDATES ENGAGE WITH THE ASSIGNMENTS?

Candidates will start by receiving a special assignment link. When a candidate opens this link, it will automatically create a private digital workspace for them on GitHub. They will then open this workspace in a code editor called Visual Studio Code to begin their work. Inside, they will find instructions and files to edit. Once they have completed the tasks and saved their files, they will use a few specific commands in the editor's terminal to submit their solution for review.

WHAT IS EXPECTED OF CANDIDATES?

Candidates are expected to engage in critical thinking and thoughtful problem-solving. Before writing any code, they are expected to analyze the assignment's requirements to structure their thoughts and design a logical plan for their solution. The goal is for candidates to apply their programming knowledge to build a well-structured and effective program that meets the evaluation standards of correctness and code quality. More importantly, candidates are expected to have a mindset of continuous improvement.

Step-by-step Instructions for Coding Assignments using GitHub

This guide outlines the step-by-step process for accessing, submitting, and receiving evaluation on coding assignments via GitHub. These assignments are designed to help you hone your programming skills. Assignments will be evaluated, but not graded or allocated marks. You are allowed up to three attempts for each assignment.

1. Setting Up

- **Create a GitHub Account**

Visit github.com and sign up using your email address to create an account¹.

- **Install Visual Studio Code (VS Code)**

Download and install VS Code from code.visualstudio.com/download, choosing the version for your operating system.

2. Accessing the Assignment

- You will receive a **GitHub Classroom assignment link** from your instructor.
- Copy and paste the link into your browser.
- Click **"Accept the Assignment"**. This will create a personal repository for you on GitHub.
- Click **"Open in Visual Studio Code"** to launch the assignment in VS Code.

3. Working on the Assignment

- In VS Code, explore the repository files (e.g., main.py, README.md, etc.).
- Carefully read the instructions provided in the README.md or assignment description.
- Edit the code as required to complete the assignment tasks.

4. Submitting Your Work

- After making your changes, save all files.
- Open the terminal in VS Code and run the following commands to submit your work:

```
bash

git add .
git commit -m "submission attempt 1"
git push origin main
```

- Replace "submission attempt 1" with "submission attempt 2" or "submission attempt 3" for subsequent attempts.

5. Evaluation Process

- Your submission will be **evaluated for correctness and code quality**.
 - **No marks or grades** will be assigned, depending on the correctness & accuracy of the codes submitted, the students will either see a green or red tick. The former (green) means coding is correct while the latter (red) means there are errors (which can be viewed in the repo under the actions tab).
- If your first attempt is not satisfactory (read red), you can **revise and resubmit** up to two more times (total three attempts).

6. Additional Notes

- **Assignments are for practice:** The focus is on learning and skill development, not on grades.
- **Use GitHub features:** You may use branches for experimentation, but ensure your final submission is on the main branch.
- **Refer to resources:** For help with Git and GitHub basics, please google and refer to GitHub's own learning resources.

Summary Table

Step	Action	Tool/Platform
Setup	Create GitHub account, install VS Code	GitHub, VS Code
Access Assignment	Accept link, open repo in VS Code	GitHub Classroom
Edit Code	Follow instructions, code in VS Code	VS Code
Submit	git add, git commit, git push	VS Code Terminal
Evaluation	Instructor reviews, feedback given	GitHub
Attempts Allowed	Up to 3	GitHub